

IoT-Based Smart Agriculture Monitoring and Automated Irrigation System

ABSTRACT

Agriculture depends heavily on environmental conditions such as soil moisture, temperature, humidity, water availability, and sunlight. Traditional irrigation methods often rely on manual observation, which can result in over-irrigation, under-irrigation, unnecessary water consumption, and delayed response to unfavorable environmental conditions.

This project presents an IoT-based smart agriculture monitoring and automated irrigation prototype using an Arduino UNO and the Wokwi simulation platform. The system continuously monitors important environmental parameters including soil moisture, temperature, humidity, ambient light intensity, and water availability. Based on predefined threshold conditions, the system automatically controls irrigation and provides visual and audible alerts.

A DHT22 sensor is used for temperature and humidity monitoring, while potentiometers are used in the simulation to represent soil moisture and water-level sensor outputs. A photoresistor is used to monitor ambient light. LEDs, a buzzer, and a 16×2 LCD are used to indicate the current system state. When the soil becomes dry and sufficient water is available, the system activates the irrigation indicator. If the water level is too low, irrigation is disabled and a warning is generated. Similarly, high humidity prevents unnecessary irrigation, while low ambient light activates a simulated grow light.

The prototype demonstrates how sensor-based monitoring and automated decision-making can improve irrigation management and reduce manual intervention. The current implementation focuses on the sensing and control layer, while cloud connectivity and remote monitoring can be incorporated as future extensions.

INTRODUCTION

The Internet of Things (IoT) provides a way to connect sensors, processing units, and actuators to create automated monitoring and control systems. In agriculture, IoT technologies can be used to collect environmental information and make decisions based on predefined conditions.

This project demonstrates a smart agriculture system using Arduino UNO as the main controller. The system collects environmental information and uses threshold-based logic to determine whether irrigation, warning alerts, or artificial lighting should be activated.

PROBLEM STATEMENT

Traditional irrigation systems often depend on manual monitoring and fixed irrigation schedules. Such systems may not respond effectively to changing environmental conditions.

The major problems include:

- Excessive water consumption due to unnecessary irrigation.
- Insufficient irrigation when soil becomes dry.
- Lack of continuous environmental monitoring.
- Delayed detection of low water availability.
- Lack of automatic response to environmental conditions.
- Dependence on manual observation.
- Difficulty in monitoring multiple parameters simultaneously.

Therefore, there is a need for a system that can continuously monitor agricultural conditions and automatically make basic irrigation and environmental control decisions.

OBJECTIVES

The main objectives of the project are:

1. To design a smart agriculture monitoring system using Arduino UNO.
2. To monitor soil moisture conditions.
3. To monitor temperature and humidity.
4. To monitor ambient light intensity.
5. To monitor water availability.
6. To automatically activate irrigation when soil is dry and sufficient water is available.
7. To prevent irrigation when the water level is too low.
8. To prevent unnecessary irrigation under high-humidity conditions.
9. To provide visual and audible alerts for abnormal conditions.
10. To display system status using a 16×2 LCD.
11. To demonstrate the complete prototype using the Wokwi simulation environment.

SCOPE OF THE PROJECT

The project focuses on developing a prototype of an automated agriculture monitoring and irrigation system.

The current scope includes:

- Environmental parameter monitoring.
- Soil moisture simulation.
- Water-level monitoring.
- Temperature and humidity monitoring.
- Ambient light monitoring.
- Automatic irrigation decision-making.
- Grow-light control.
- Visual LED indicators.
- Audible alerts.
- LCD-based status display.
- Serial-monitor-based monitoring.
- Simulation using Wokwi.

The current prototype does not include physical water pumping hardware, cloud storage, mobile application integration, or remote Internet-based monitoring. These features can be added in future versions.

LITERATURE / TECHNOLOGY BACKGROUND

Internet of Things in Agriculture

The Internet of Things refers to the connection of physical devices containing sensors, processing units, and communication capabilities so that information can be collected and processed automatically.

In agriculture, IoT systems can monitor parameters such as:

- Soil moisture
- Temperature
- Humidity
- Light intensity
- Water level
- Weather conditions

Smart Agriculture

Smart agriculture uses technology and data to improve agricultural productivity and resource utilization.

Instead of relying entirely on manual observation, smart agriculture systems use sensors and automated decision-making to monitor field conditions.

The main advantages include:

- Better resource utilization.
- Reduced manual effort.
- Real-time monitoring.
- Improved irrigation management.
- Faster response to environmental changes.

Arduino UNO

Arduino UNO is a microcontroller development board based on the ATmega328P microcontroller.

It provides:

- Digital input/output pins.
- Analog input pins.
- USB programming interface.
- Serial communication.
- Power supply connections.
- PWM-supported digital pins.

In this project, Arduino UNO acts as the central controller. It receives readings from the sensors, processes them using threshold-based conditions, and controls the LEDs, buzzer, LCD, and irrigation indicator.

DHT22 Sensor

The DHT22 is a digital sensor used to measure temperature and relative humidity.

In this project, the DHT22 provides:

- Temperature in degrees Celsius.
- Relative humidity in percentage.

The Arduino reads these values and uses them as part of the irrigation and environmental decision-making process.

Soil Moisture Monitoring

Soil moisture indicates the amount of water available in the soil.

In the physical system, a soil moisture sensor can provide an analog signal corresponding to soil conditions.

In this Wokwi prototype, a potentiometer connected to analog pin A0 is used to simulate the soil moisture sensor output.

A lower simulated value represents drier soil.

Water-Level Monitoring

Water availability is important because the irrigation system should not attempt to operate when sufficient water is unavailable.

A potentiometer connected to analog pin A2 is used in the Wokwi prototype to simulate a water-level sensor.

A lower value represents a low water condition.

Photoresistor

A photoresistor, also called an LDR, is a light-sensitive component whose resistance changes according to the amount of light falling on it. The system uses the photoresistor to determine ambient light intensity. The measured value is converted to an approximate lux value. If the ambient light falls below the defined threshold, the system activates the yellow grow-light indicator.

SYSTEM ARCHITECTURE

The system consists of three major layers:

Input Layer

The input layer contains the sensors used to collect environmental information.

- DHT22
- Soil moisture simulation
- Water-level simulation
- Photoresistor

Processing Layer

The Arduino UNO acts as the processing and control unit.

1. Reads sensor values.
2. Compares readings with predefined thresholds.
3. Determines the current environmental condition.
4. Controls the required outputs.

Output Layer

The output layer provides information and control through:

- Blue LED for irrigation/pump indication.
- Green LED for normal/safe condition.
- Red LED for low-water warning.
- Yellow LED for grow-light indication.
- Buzzer for alerts.
- 16×2 LCD for status display.
- Serial Monitor for detailed readings.

WORKING PRINCIPLE

The system continuously reads environmental parameters and evaluates them using predefined thresholds.

The main thresholds are:

Parameter	Threshold
Soil moisture	400
Water level	300
Humidity	80%
Temperature	35°C
Ambient light	300 lux

The system follows a priority-based decision process.

Case 1: Dry Soil + Sufficient Water

If the soil value is below 400 and water availability is above 300, the system considers the soil dry.

If humidity is also below 80%, irrigation is activated.

Output:

- Blue LED ON
- Green LED ON
- Pump indicator ON

Case 2: Low Water

If the water-level value falls below 300, irrigation is disabled regardless of soil condition.

Output:

- Pump OFF
- Red LED ON
- Buzzer ON

This prevents the system from attempting irrigation when water availability is insufficient.

Case 3: Dry Soil + High Humidity

If soil is dry but humidity is 80% or higher, irrigation is not activated.

This prevents unnecessary watering when atmospheric humidity is already high.

Output:

- Pump OFF
- Green LED ON
- LCD shows high humidity condition

Case 4: Temperature $\geq 35^{\circ}\text{C}$

If temperature reaches or exceeds 35°C , the system activates the buzzer as an environmental warning.

Output:

- Buzzer ON
- High-temperature status displayed

The temperature condition is monitored independently from the irrigation condition.

Case 5: Light < 300 Lux

If the measured ambient light is below 300 lux, the system assumes that the environment is relatively dark.

The yellow LED is activated as a simulated grow light.

Output:

- Yellow LED ON
- LCD shows low-light condition

Case 6: Normal Conditions

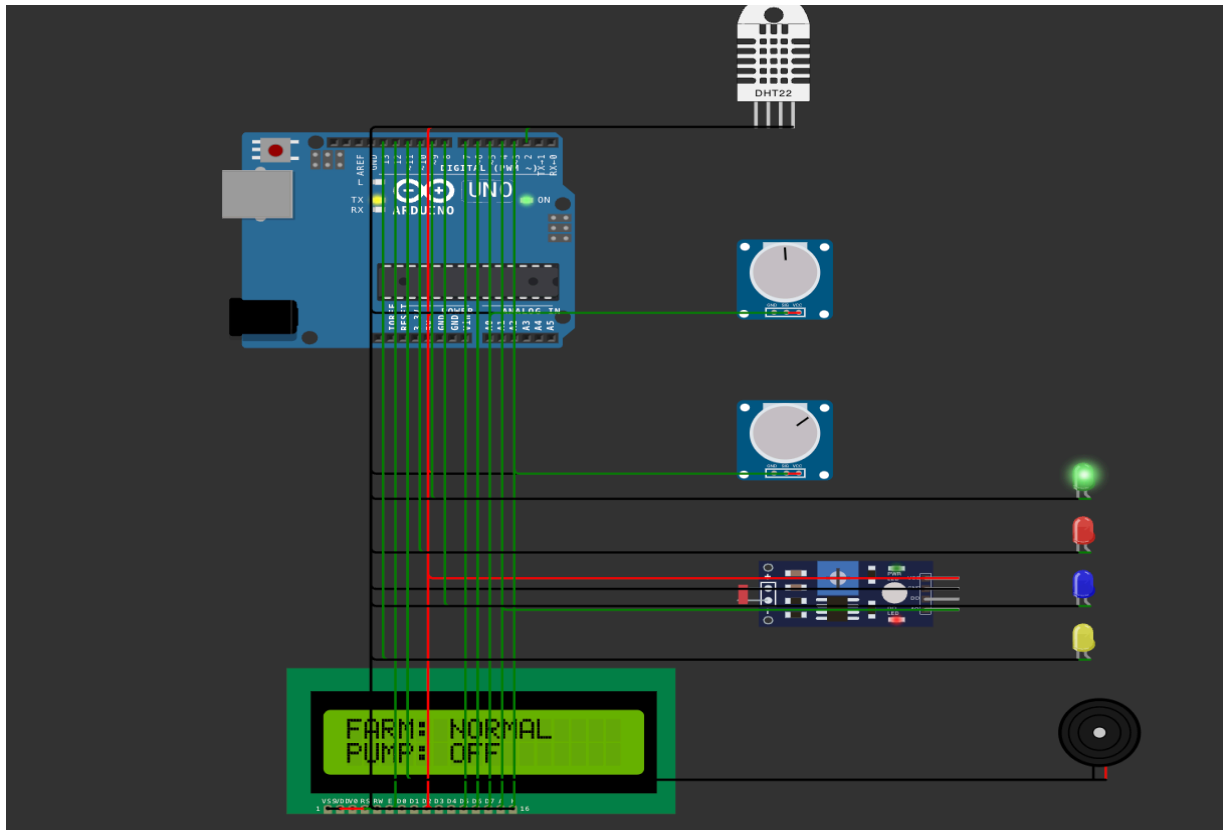
If none of the abnormal conditions are active, the system remains in normal mode.

Output:

- Green LED ON
- Pump OFF

- Red LED OFF
- Buzzer OFF
- Grow light OFF

Circuit Design and Working



The circuit consists of an Arduino UNO connected to environmental sensors and output devices. The DHT22 is connected to digital pin D2 for temperature and humidity monitoring. Analog pin A0 receives the simulated soil moisture value, while A2 receives the simulated water-level value. The photoresistor is connected to A1 for ambient light monitoring.

The LCD is connected to digital pins D12, D7, D6, D5, D4, and D3. LEDs are connected to digital pins D8, D9, D10, and D13 to indicate irrigation, normal operation, low water, and grow-light conditions respectively. The buzzer is connected to D11 to provide audible alerts.

Working

The system continuously reads the environmental parameters through the connected sensors. The soil moisture value is used to determine whether irrigation is required. When the soil is dry and sufficient water is available, the blue LED is activated to represent the irrigation pump. The water-level reading is used to prevent irrigation when the available water is too low. In this situation, the pump remains OFF and the red LED and buzzer provide an alert. Humidity is also considered before irrigation. If the soil is dry but humidity is already high, irrigation is prevented to avoid unnecessary watering. The DHT22 continuously monitors temperature. When the temperature reaches or exceeds 35°C, the buzzer provides an alert. The photoresistor measures ambient light. When the light intensity falls below 300 lux, the yellow LED is activated to represent a grow light. The LCD provides a simple local display of the current system status, while the Serial Monitor provides detailed sensor readings.

Advantages, Limitations and Future Scope

Advantages

- The system provides automated irrigation based on environmental conditions and reduces the need for continuous manual observation.
- It can monitor multiple parameters simultaneously, including soil moisture, water availability, temperature, humidity, and ambient light.
- The system also provides immediate visual and audible alerts, making abnormal conditions easier to identify.
- The Arduino-based implementation is relatively simple and can be expanded with additional sensors and communication modules.

Limitations

- The current implementation is a Wokwi-based simulation rather than a physical agricultural installation.
- Potentiometers are used to simulate the analog outputs of soil moisture and water-level sensors.
- The blue LED represents the irrigation pump instead of controlling an actual water pump.
- The system does not currently provide cloud connectivity or remote mobile monitoring.

- The threshold values are fixed and may need to be changed depending on crop type, soil conditions, climate, and agricultural requirements.

Future Scope

The prototype can be extended by integrating Wi-Fi or other communication technologies for remote monitoring.

A mobile or web application could be developed to allow farmers to view sensor values and system status remotely.

A real capacitive soil moisture sensor, water-level sensor, relay module, and water pump could replace the simulated components.

Cloud storage could be added to maintain historical sensor data and generate graphs.

Machine learning could also be incorporated to predict irrigation requirements based on historical environmental conditions, weather information, crop type, and soil characteristics.

Solar power could be used to operate the system in remote agricultural areas.

Conclusion

The IoT-Based Smart Agriculture Monitoring and Automated Irrigation System demonstrates how Arduino, sensors, and automated control can be applied to agricultural monitoring.

The system monitors important environmental parameters including soil moisture, temperature, humidity, water availability, and ambient light. Based on these parameters, it provides appropriate responses such as activating the irrigation indicator, generating warnings, or activating a simulated grow light.

The Wokwi prototype successfully demonstrates the sensing, processing, and output stages of the system without requiring physical hardware. The use of multiple environmental parameters allows the system to make more informed decisions than a simple fixed irrigation system.

Although the current prototype uses simulated soil and water sensors and does not include Internet connectivity, it provides a foundation that can be extended into a

complete IoT agriculture solution using real sensors, pumps, Wi-Fi connectivity, cloud services, and mobile applications.

The project therefore demonstrates the potential of embedded systems and IoT technologies for reducing manual intervention and improving resource management in agriculture.

Output

Output 1

```
=====
      SMART FARM STATUS
=====
Soil Moisture : 500
Temperature   : 28.0 C
Humidity      : 60.0 %
Light         : 500.8 lux
Water Level   : 700
Irrigation    : OFF
Grow Light    : OFF
STATUS        : CONDITIONS NORMAL
=====
```

Output 2

```
=====
      SMART FARM STATUS
=====
Soil Moisture : 114
Temperature   : 28.0 C
Humidity      : 60.0 %
Light         : 500.8 lux
Water Level   : 700
Irrigation    : ON
Grow Light    : OFF
STATUS        : SOIL DRY - IRRIGATING
=====
```